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Olympic Insights Using Data Analytics

AN INTERACTIVE POWER BI DASHBOARD

Power BI
Data Visualization

Excel / CSV
Data Sources

DAX
Advanced Analytics

Presented by
Chetan Agrawal

Course
Data Analytics

Date
09-10-2026



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Project Overview & Problem Statement

From Past Performance to Future Possibilities



Project Overview

- Analyze historical Olympic data to gain insights into athlete performance, country-wise medal trends and sport-specific dominance.
- Track achievements and evaluate performance KPIs across Olympic Games.
- Identify key trends in country participation, athlete demographics and sports popularity.
- Build an interactive Power BI dashboard to provide data-driven insights for strategic planning and athlete development.

Problem Statement

The International Olympic Committee (IOC) needs advanced insights into athlete performance, country-wise medal trends, and sport-specific dominance across Olympic Games.

Key Objectives

- Understand historical Olympic performance.
- Analyze country-wise medal trends.
- Identify sport-specific dominance.
- Explore athlete participation and demographics.
- Support strategic planning and development with data-driven insights.

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Dataset & Data Preparation

CLEAN DATA | RELIABLE INSIGHTS | BETTER DECISIONS

Quality Data Builds Stronger Insights

Datasets Used

Olympics_Country.csv	Country details and related information
Olympics_Games.csv	Games and event information
Olympic_Athlete_Bio.csv	Athlete personal and demographic details
Olympic_Athlete_Event_Results.csv	Athlete participation and event results
Olympic_Games_Medal_Tally.csv	Medal tally by country and year
Olympic_Results.csv	Detailed results and medal information

Data Preparation Process

1 Data Import

- Loaded all 6 datasets into Power BI.
- Checked file structure and initial data preview.

2 Data Cleaning

- Handled missing values (if any).
- Removed duplicates (where applicable).
- Standardized data formats and text values.

3 Data Transformation

- Renamed columns for consistency.
- Set correct data types.
- Created calculated fields (if required).

4 Data Modelling

- Built relationships between tables.
- Configured star schema model.
- Ensured proper filter context.

The prepared dataset is used to create **6 interactive dashboards** in **Power BI**, enabling in-depth analysis of athlete performance, country-wise medal trends, sports dominance, and demographic insights.



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Tools, Methodology & KPIs

POWER BI | DATA PROCESSING | DAX | INTERACTIVE VISUALS

Data Driven Insights for a Stronger Olympic Tomorrow



Tools Used

Power BI Desktop
Dashboard development & visualization

Excel / CSV
Data source files

Power Query
Data transformation & cleaning

DAX
Calculated measures & KPIs

Methodology / Workflow

Data Sources
Importing CSV files

Data Cleaning
Handling missing values & duplicates

Data Modelling
Creating relationships and structure

DAX Measures
Building calculated measures & KPIs

Visualizations
Charts, tables, filters & slicers

Insights
Interactive dashboard with 6 pages

Key Performance Indicators (KPIs)

- **Total Medals Awarded** (Gold + Silver + Bronze)
- **Total Countries Participated**
- **Total Athletes Represented**
- **Total Events Held**
- **Top 10 Countries** by Total Medals
- **Medal Trends** Over the Years

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Global Medal Overview

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
- Sport & Event Analysis
- Host Country Analysis
- Gender & Demographics

Filters

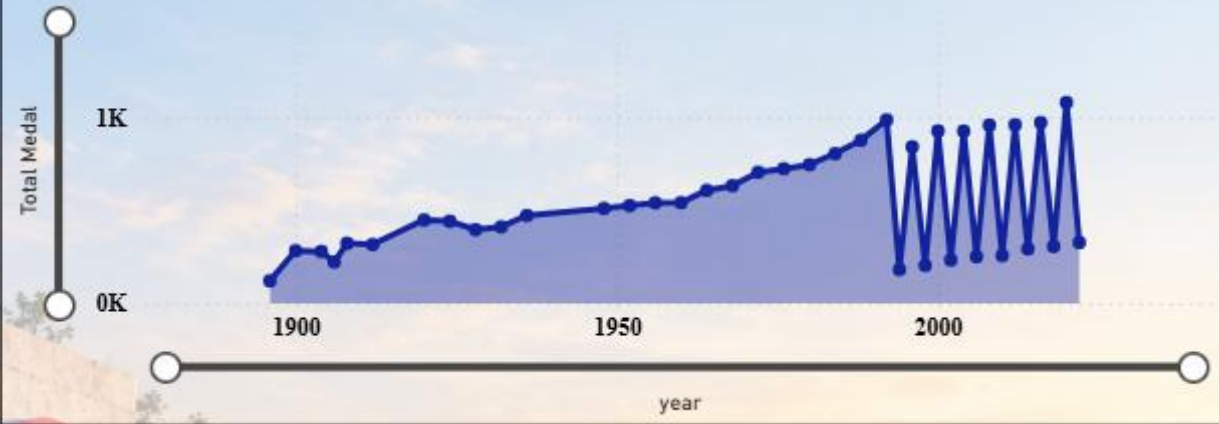
Country
All

Year
All

Sex
All

Total_Medal	Total_Event_Held	Total_Country	Total_Athlete
21K	964	235	156K

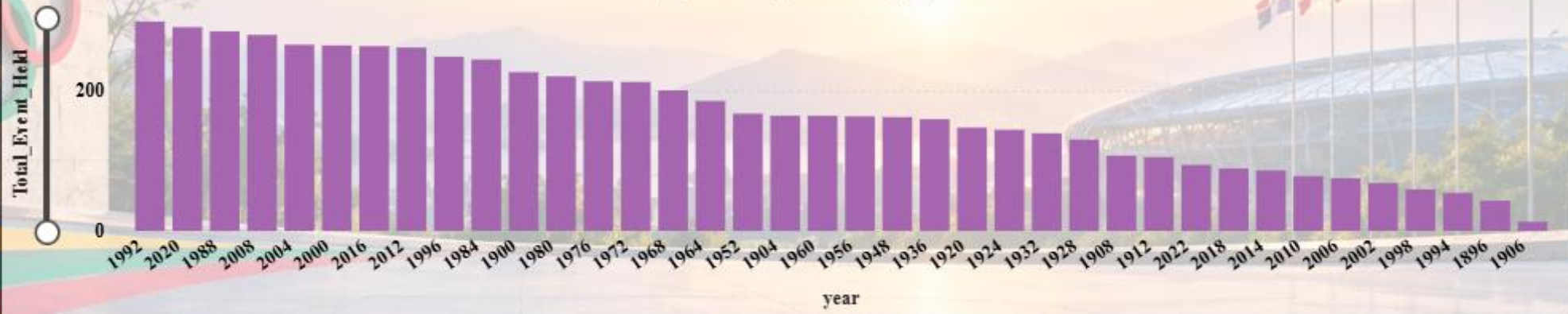
Medal Trends Over the Years



Country Wise Total Medal

country	Total Medal
United States	3009
Soviet Union	1204
Germany	1098
Great Britain	988
France	951
Italy	789
People's Republic of China	713
Total	10593

Total_Event_Held by year



Country Insights

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
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Filters

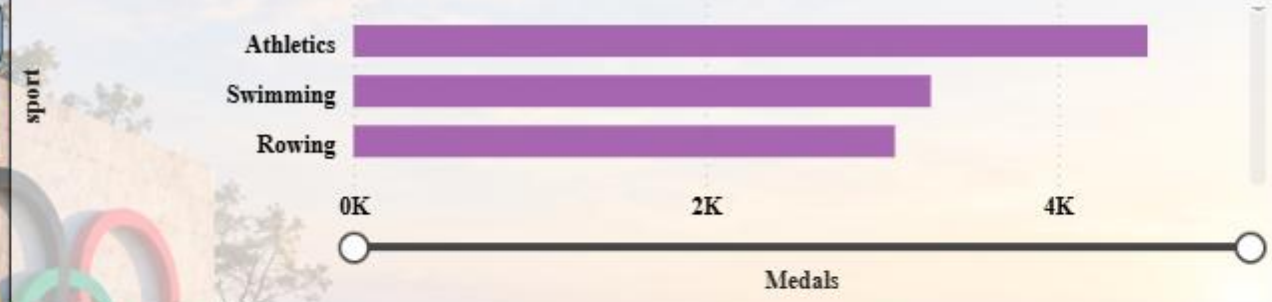
Country
All

Year
All

Sex
All

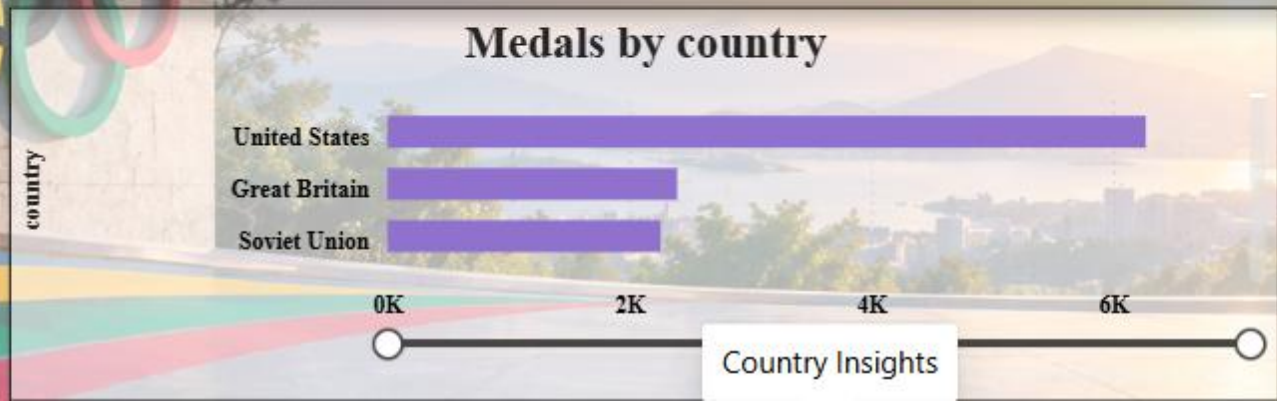
156K	Total_Athlete
235	Total_Country
13	Medal_Efficiency
964	Total_Event_Held
21K	Total_Medal
7K	Total_Gold_Medal
7K	Total_Silver_Medal
7K	Total_Bronze_Medal

Medals by sport

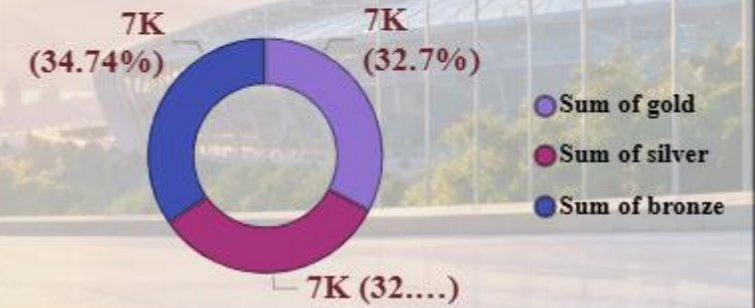


country	Total Medal	year
Australia	2	1896
Austria	5	1896
Denmark	6	1896
France	11	1896
Germany	13	1896
Great Britain	7	1896
Total	20654	

Medals by country



Medal Distribution



Country Insights

Athlete Spotlight

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
- Sport & Event Analysis
- Host Country Analysis
- Gender & Demographics

Filters

Country
All

Year
All

Sex
All

156K
Total_Athlete



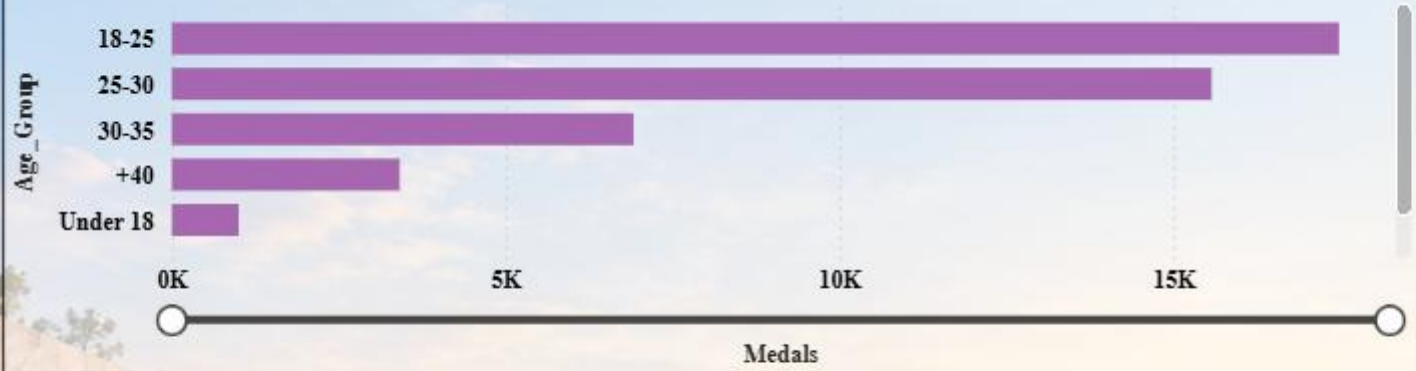
Ioannis Theofilakis
Top_Athlete



21K
Total_Medal



Medals by Age_Group



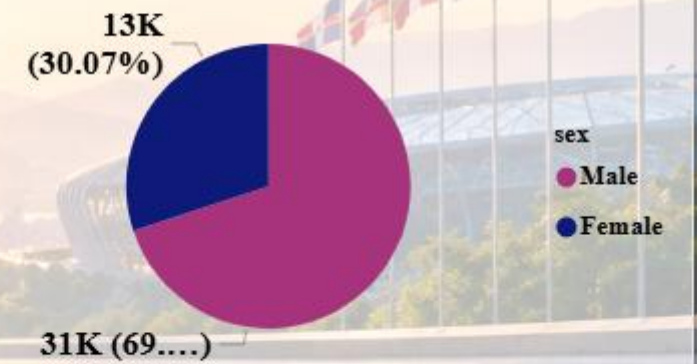
athlete	Total_Event_1
Abdul Rashid	4
Abel	3
Abel Braga	1
Abelardo	1
Abner	1
Abram	4
Acedo	1
Total	

athlete	Medals
Boris Shakhlin	13
Edoardo Mangiarotti	13
Ireen Wüst	13
Jenny Thompson	13
Larisa Latynina	18
Marit Bjørgen	15
Michael Phelps	28
Nikolay Andrianov	15
Total	154

Total_Athlete and Medal_Winning_Athlete



Medals by sex



Athlete Spotlight

Sport & Event Analysis

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
- Sport & Event Analysis
- Host Country Analysis
- Gender & Demographics

Filters

Country: All

Year: All

Sex: All

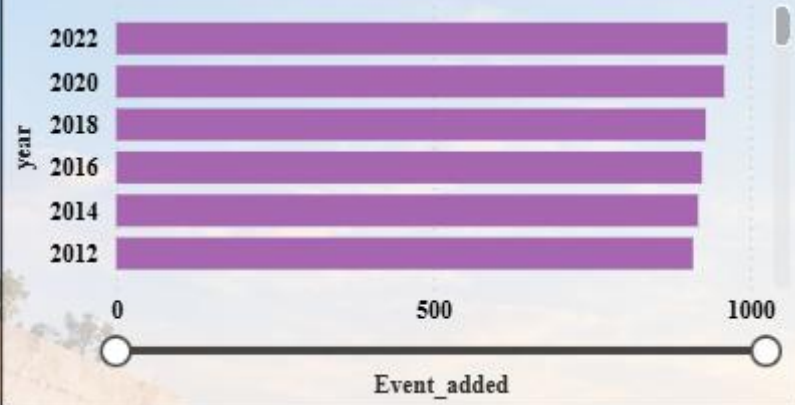
156K Total_Athlete

964 Total_Event_Held

Athletics Top_Sport_By_Athlete

Athletics Top_Sport_By_Medal

Event_added by year

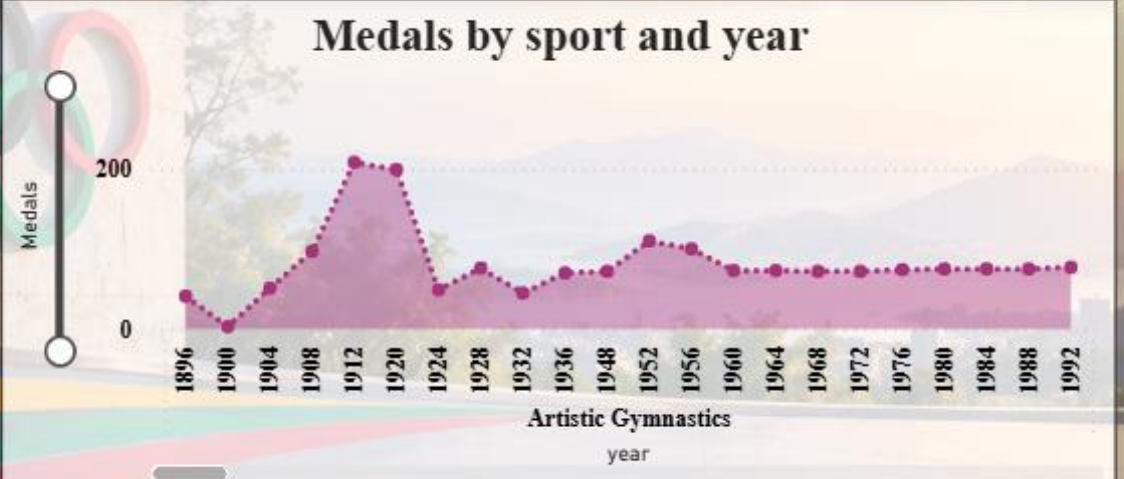


Total_Event_Held by year



country	Count of sport
	4
Afghanistan	9
Albania	9
Albania Australia	1
Albania Bulgaria	1
Albania Greece	1
Albania Kosovo	1
Albania United States	1
Total	112

Medals by sport and year



sport	Total_ Athlete
3x3 Basketball	64
Aeronautics	1
Alpine Skiing	3118
Alpinism	25
American Football	213
Archery	1190
Art Competitions	1035
Total	155867

sport	Medals
Wrestling	1356
Weightlifting	678
Water Polo	1180
Volleyball	1080
Tug-Of-War	98
Triathlon	48
Trampolining	36
Tennis	358
Total	44687

Host Country Analysis

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
- Sport & Event Analysis
- Host Country Analysis
- Gender & Demographics

Filters

Country
All

Year
All

Sex
All

156K
Total_Athlete



964
Total_Event_Held



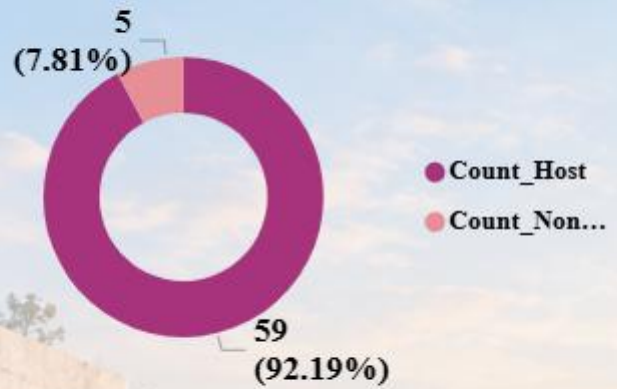
59
Count_Host



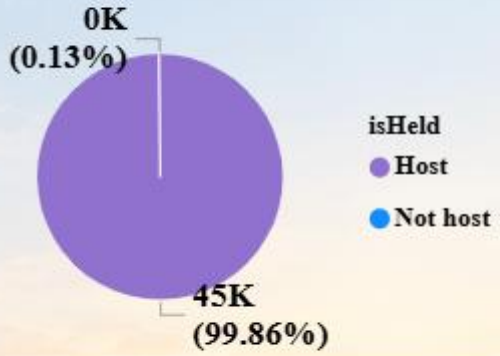
235
Total_Country



Count_Host and Count_NonHost

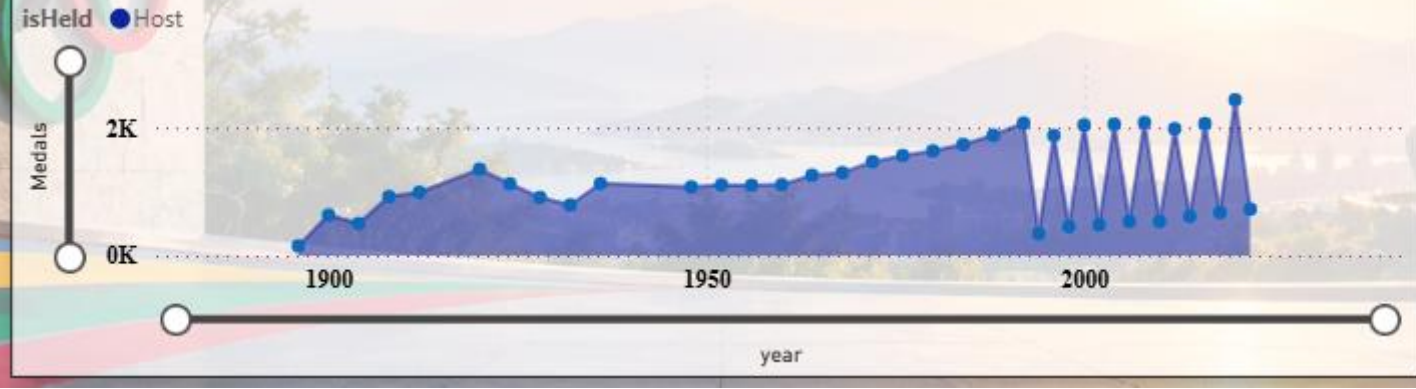


Medals, Count_Host and Count_NonHost by isHeld



country_noc	year	Total_Event_Held
GRE	1896	43
FRA	1900	226
USA	1904	164
GRE	1906	13
GBR	1908	107
SWE	1912	105
BEL	1920	147
FRA	1924	144
Total		964

Medals by year and isHeld



country_noc	Count of year
AUS	3
AUT	2
BEL	1
BRA	1
CAN	3
CHN	2
ESP	1
FIN	2
Total	64

Gender & Demographics

Pages

- Global Medal Overview
- Country Insights
- Athlete Spotlight
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- Gender & Demographics

Filters

Country
All

Year
All

Sex
All

116K
Male



40K
Female



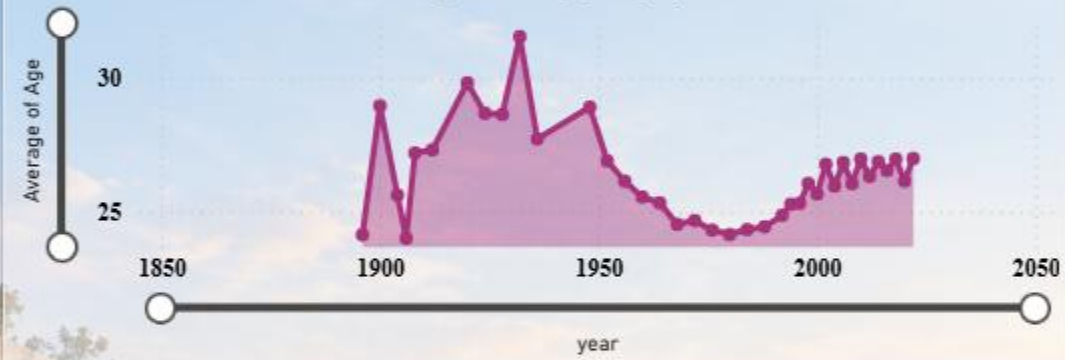
Hend Zaza,Di
Young_Winner



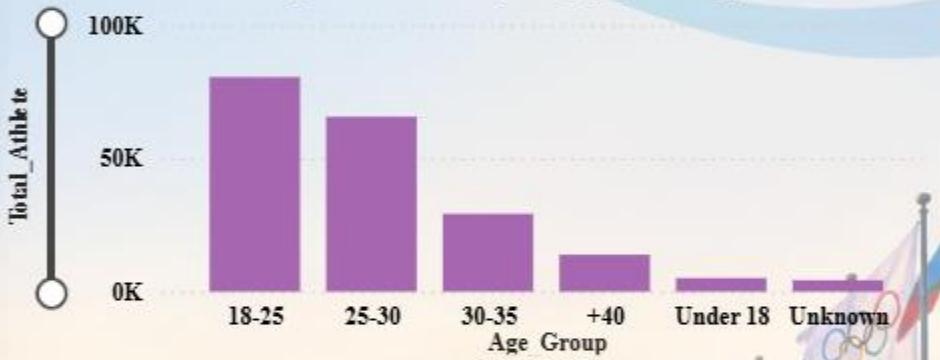
John Quincy
Oldest_Winner



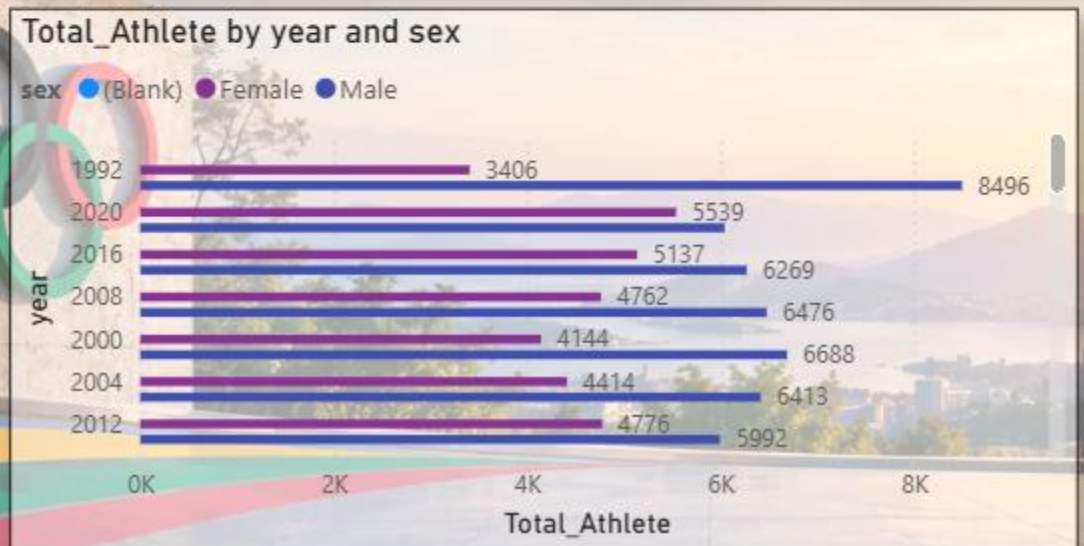
Average of Age by year



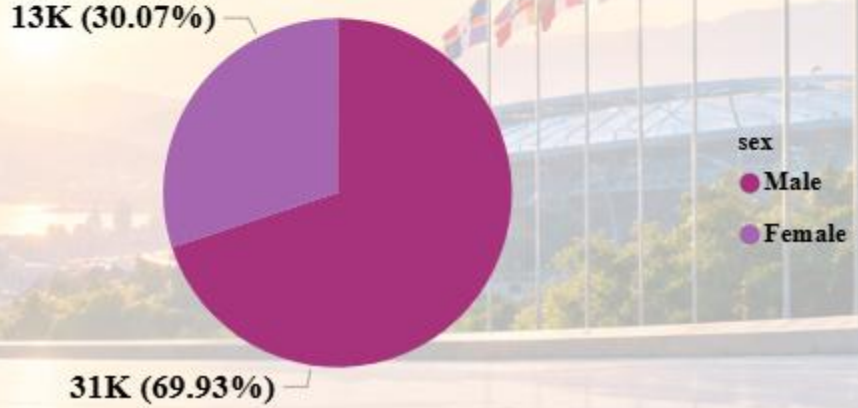
Total_Athlete by Age_Group



Total_Athlete by year and sex



Medals by sex



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Key Insights & Findings

More Insights Bigger Impact

DATA-DRIVEN INSIGHTS FOR A STRONGER OLYMPIC TOMORROW

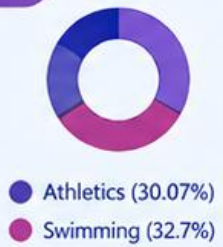
1. Historical Medal Leaders

- United States leads in total medals with 3,098 medals.
- Soviet Union follows with 1,204 medals.
- Germany ranks third with 1,098 medals.



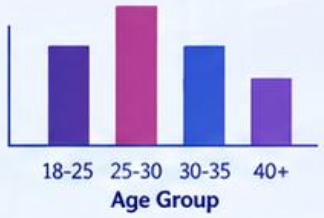
2. Sports with Most Medals

- Athletics is the top sport with 13K (30.07%) medals.
- Followed by Swimming with 7K (32.7%).
- Other sports also contribute significantly to the total medal count.



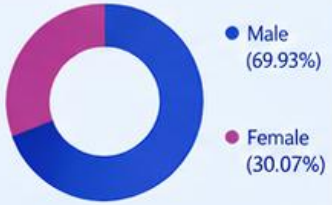
3. Athlete Age & Participation

- Majority of medal winners are in the 20–30 age group.
- Participation ranges from young athletes to experienced veterans.



4. Gender Participation

- Male athletes outnumber female athletes in total participation.
- Female participation has shown steady growth over the years.
- Gender-wise medal distribution is more balanced in recent editions.



5. Host Country Trends

- Host countries have a strong presence in medal counts.
- Medal trend shows fluctuations across different editions.
- Hosting impacts event participation and overall performance.



6. Overall Takeaways

- The USA remains the most successful nation in Olympic history.
- Athletics and Swimming are the most dominant sports.
- Younger athletes (20–30) and male participation are higher, though female participation is growing.
- Host countries show strong but variable medal performance over time.

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Conclusion & Future Scope

Better Insights for a Stronger Olympic Future

KEY TAKEAWAYS | IMPACT | NEXT STEPS

Conclusion

- The dashboard provides a comprehensive view of Olympic history, covering medals, athletes, sports, countries, and host nations.
- Data-driven insights help identify performance trends, dominance patterns and participation growth across different dimensions.
- The interactive dashboards enable easy exploration and better decision-making for strategic planning and athlete development.
- The project demonstrates the power of Power BI in transforming complex Olympic data into meaningful visual insights.

Key Takeaways

Medal Leaders
Identify top-performing countries and historical trends.

Sports Dominance
Understand which sports contribute most to medals and participation.

Athlete Insights
Explore age, gender and performance patterns of athletes.

Host & Demographics
Analyze the impact of host countries and gender/age distribution.

Future Scope

Medal Trend Forecasting
Predict future medal trends using advanced analytics and ML techniques.

Next Edition Predictions
Estimate potential performance for upcoming Olympic Games.

Deep Athlete Performance Analysis
Include advanced metrics like win rate, consistency and event-wise performance.

Extended Data Integration
Combine with external datasets (e.g., population, GDP, sports investment).

Thank You!

ANY QUESTIONS?

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